

A Comparative Study of Gaussian Processes and Bayesian Additive Regression Trees for Stock Return Prediction

Atticus Roop

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Abstract

This project studies the applications Gaussian Processes and Bayesian Additive Regression Trees to stock market data. The primary objective is to evaluate whether these Bayesian methods can address limitations of traditional time series models when applied to financial data. Historical market data is pulled from the yahoo finance package in R (yahoofinancer package, n.d.). The analysis revolves around finding the most consistent period of returns that the model can predict then engineering features to predict volatility, momentum and other market dynamics. Although both approaches are grounded in Bayesian principles, they differ substantially in implementation. Bayesian Additive Regression Trees rely on sets of decision trees made up of smaller regressions, when combined they create a powerful predictive tool. Gaussian Processes model data through probabilistic covariance structures assigning probabilities to potential outcomes. The analysis targeted predicting daily, monthly, quarterly and yearly returns, results indicated that short-term returns exhibit a high degree of randomness and are difficult to predict with either method. However when tested on longer periods of time Gaussian Processes and Bayesian Additive Regression Trees outperformed their counterparts such as linear models and A Random Walk (Malkiel, 1973).

1 Introduction

Financial markets are plagued by uncertainty and unpredictability, prices not only reflect all available information but also the fears of investors and their expectations. The unpredictability in asset prices reflects a popular market theory that stems from A Random Walk Down Wallstreet, by Burton Malkiel (Malkiel, 1973). His theory states that stock markets are impossible to predict and the movement in prices is essentially random. Malkiel believes that tools cannot be created to predict the market and your best course of action is to invest in the market as a whole. This paper explores the implementation of Gaussian Processes and Bayesian Additive Regression Trees in financial data. Over the past decade, Bayesian models have gained popularity in the finance community (Jacquier & Polson, 2010; Miao et al., 2015). Their applications to risk management, options trading and market making has made them a valuable tool. Unlike simple linear regression's, Bayesian models provide a probabilistic framework that captures the full distribution of possible outcomes rather than a single point estimate (van de Schoot et al., 2021; Mauboussin & Callahan, 2025). In simple linear regressions our dependent variable is predicted through our coefficients(β term). For each different value and combinations of values of our independent variables on our beta terms we create a single estimate for our dependent variable. Our uncertainty is only reflected through our error term. Simple linear regression does not account for enough uncertainty which makes them a poor fit for financial data. In contrast, Bayesian models treat β as a random variable, they are assigned a range of outcomes rather than one single value. There are many Bayesian methods, based on the times series aspect of financial data Gaussian Processes and Bayesian Additive Regression Trees were determined to be best suited.

1.1 Gaussian Processes

Gaussian Processes are especially useful for the time series structure of financial data because they model dependence between observations through a covariance function, also known as a kernel (Roberts et al., 2013). Instead of assuming that each data point is independent, the kernel measures how closely related points are across time. This is important because market data often contains patterns where

recent observations carry more information than older ones, especially during periods of changing volatility or momentum. A Gaussian Process places a prior distribution over an unknown function, allowing the model to consider many possible price paths rather than forcing the data into one fixed equation (Roberts et al., 2013; Gonzalez et al., 2019). These possible paths are represented through a multivariate normal distribution, where multiple points in time are modeled jointly. As a result, Gaussian Processes provide both a prediction and a measure of uncertainty, making them well suited for noisy, nonlinear financial data where relationships can shift over time.

1.2 Bayesian Additive Regression Trees

The second Bayesian approach that will be analyzed are Bayesian additive regression trees. Unlike Gaussian Processes, Bayesian Additive Regression Trees use a sum-of-trees model to approximate a unknown function (Chipman et al., 2010). Bayesian additive regression tree models do not rely on a complex model, instead they rely on many smaller simple trees, where each tree captures a small part of the overall pattern in the data. When these trees are added together, they form a flexible prediction function, with an error term accounting for the remaining unexplained variation. This structure is useful for financial data because asset prices and returns often move in nonlinear ways. A single regression model may struggle to capture sudden changes in momentum, volatility, or market behavior. Models that follow a similar framework to BART models often run into an overfitting problem. BART compensates for this by its regularization prior to computation (Chipman et al., 2010). No one singular regression will dominate the model. The Bayesian aspect of BART models stems from all components of the model being treated as random variables. Inference is conducted using Markov Chain Monte Carlo methods, which generate a posterior distribution over possible functions. This allows us to not only estimate one prediction but also the uncertainty surrounding the prediction. A BART model and a GP model will allow for two different approaches to handling volatile data. BART models are better suited to handle sharp, non-linear changes in asset prices while GP models impose smoothness when projecting the data due to the covariance kernel used.

2 Data

Historical stock market data is retrieved from the `yahoofinancer` package in R (`yahoofinancer` package, n.d.). This package allows you to pull historical data on any ticker, for the research the ticker focused on is VOO. VOO is the Vanguard ETF which is designed to track the performance of the S&P 500. The S&P 500 is the market index that tracks the performance of the 500 leading companies listed on the stock exchange. It represents approximately 80% of the total market capitalization of U.S. companies. It is widely considered one of the best indicators of overall U.S stock market performance. The data being pulled from the VOO ticker consists of the variables, Date:Closing Price:adj close:Opening Price:Daily High:Daily Low:Volume. Adjusted close is a stock's closing price modified to account for corporate actions like dividends, stock splits, and rights offerings.

2.1 Exploratory Data Analysis

Understanding the initial trends and movement in prices, returns, volume is crucial to properly apply a predictive model to the S&P 500 data. We first must look at the initial shape in closing price as this will give us an idea about what kind of transformations need to be applied.

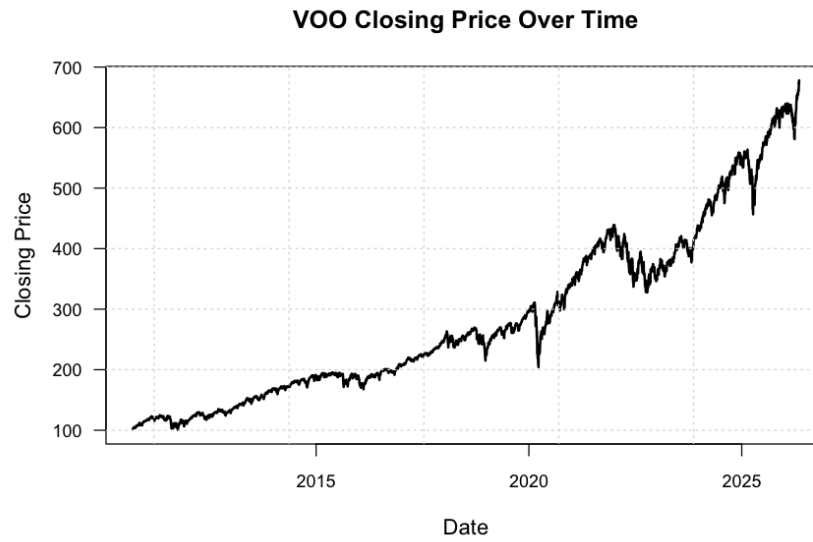


Figure 1: VOO Closing Price Over Time

The plot of the closing price shows the long-term trend in the S&P 500. The closing price is non-stationary trending positively over the examined period. A popular approach to modeling stocks is to use the returns instead of the closing price (Jacquier & Polson, 2010).

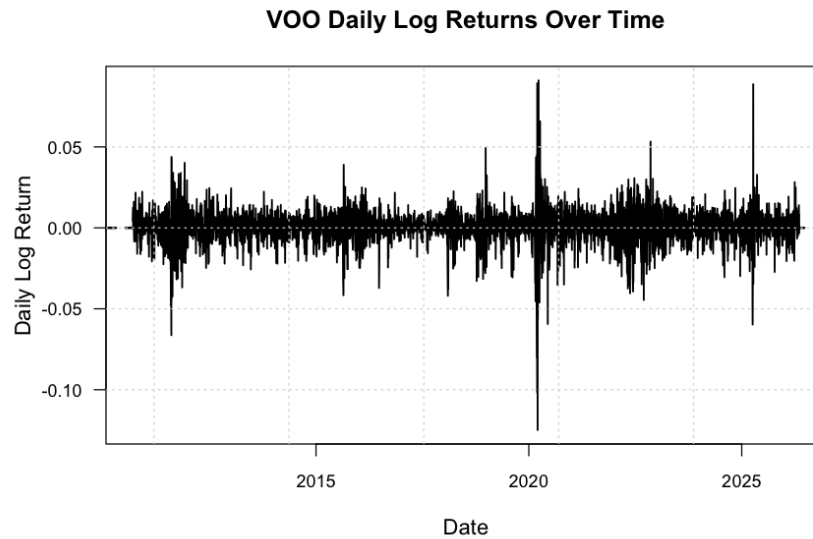


Figure 2: VOO Daily Returns

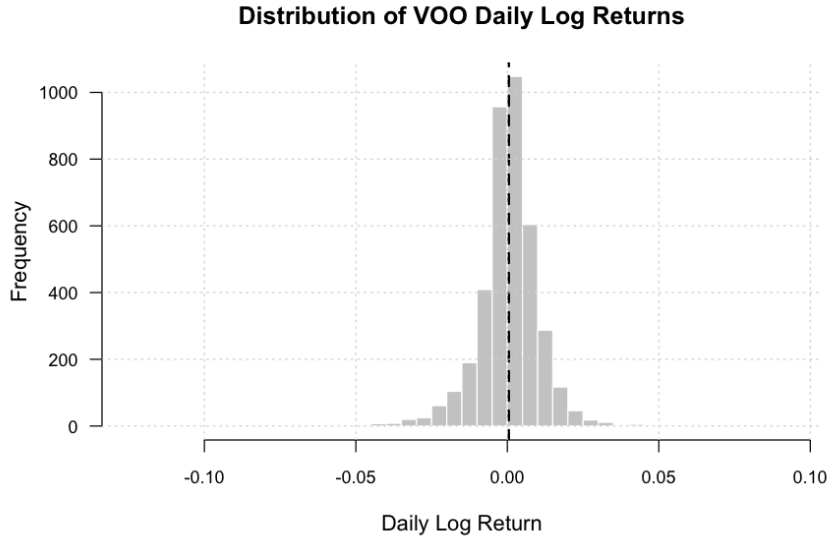


Figure 3: VOO Daily Returns Distribution

Daily returns oscillate around zero showing much less trend than the closing price plot. Stationary data is much easier to model because it represents purely daily market movement. The histogram shows the majority of daily returns are concentrated around zero, extreme daily movements do still occur, however, they are rare.

3 Literature Review

Prior research on Gaussian Processes has emphasized their usefulness for time-series data because they provide a probabilistic framework for modeling uncertainty, dependence, and changing patterns over time. Roberts et al. (2013) describe Gaussian Processes as a Bayesian non-parametric approach that is especially well suited for time-series modeling because the model does not force the data into one fixed functional form. Roberts et al. (2013) explain that covariance is the “heart” of Gaussian Process inference because it determines how observations are related to one another across time. In a time-series setting, the kernel measures the relationship between different points in time and organizes these relationships into a covariance matrix. This allows the model to use nearby observations to inform predictions while still allowing uncertainty to increase when the model moves farther away from observed data.

Financial research has also begun to apply Gaussian Processes more directly to investment problems. Gonzalez, Lezmi, Roncalli, and Xu (2019) examine financial applications of Gaussian Processes and Bayesian optimization, noting that Gaussian Processes are connected to kernel methods and have developed alongside machine learning approaches in finance. Their work applies Gaussian Process regression to financial problems such as interest-rate modeling and trend-following strategies. They support the idea that Gaussian Processes do have applications in financial settings because GP’s can model smooth patterns, estimate uncertainty and can be compared to popular simpler benchmarks like *A Random Walk*. The literature also emphasizes that the choice of kernel strongly affects the behavior of the Gaussian Process. Gonzalez et al. (2019) explain that Gaussian Processes can be viewed as probability distributions over functions, meaning the covariance function controls the properties of the functions being modeled. Different kernels can impose different assumptions about smoothness, periodicity, monotonicity, and overall structure. Explored in this paper is the Matérn 5/2 kernel which is useful for relationships that are smooth, but not perfectly smooth. Another common kernel used in Gaussian Process models is the squared exponential kernel. The squared exponential kernel assumes that the underlying function is very smooth which would not be very useful for financial data. Overall, prior research suggests that Gaussian Processes are useful for time-series prediction because they directly model uncertainty and dependence across observations. However, the same literature also

implies a challenge for financial forecasting as stock returns may not always follow smooth patterns.

Past research on Bayesian Additive Regression Trees and their direct application to financial data remains relatively limited. However, recent studies have begun to explore how Bayesian tree-based methods can be used to model asset returns and portfolio behavior. For example, in *Bayesian Portfolio Sorts*, Creal and coauthors apply a sum-of-trees framework to excess returns, using multiple independent regression trees to capture complex relationships in financial data (Creal & Kim, 2021). This is relevant to stock return prediction because asset returns may be driven by several different market conditions at once, including volatility, momentum and changing risk. However, the literature also suggests that BART models must be used carefully. Their flexibility can be an advantage when modeling nonlinear return behavior, but financial data often contains noise that may appear predictive in a training sample but disappear in testing. This makes model evaluation especially important. This paper compares multiple horizons trying to avoid this overfitting problem.

4 Feature Engineering

The variables provided by the dataset don't provide much information for a model to predict the future stock price accurately. Stock prices are noisy and difficult to predict, the goal of feature engineering will be to create variables that capture many distinct features of stock price movement. The first feature added to the dataset is lagged returns, these features test whether past market movements contain information about future market returns. The variables t-1 through t-30 are included in the dataset to represent the return from 1 past trading day to 30 past trading days before the current observation. If the lagged returns have predictive power it would suggest that current price movements have short-term memory.

Multi-horizon returns are the next feature to be added to the dataset. This includes periods such as 3-day, 5-day, 20-day and 252-days. These periods represent targeted return windows which are important in modeling financial data. Daily, Weekly, Quarterly and Yearly are target periods for the models we will test. Next volatility ratios compare the short-term and long-term volatility. One example in the context of this project is a ratio of 5-day volatility to 20-day volatility which can show if recent risk is higher than historical risk. Very similar to volatility volatility ratios, rolling volatility is engineered to measure how market risk has changed over time. These variables measure how much volatility has changed in 5, 20, 60 or 120 day windows. Finally a few advanced features were engineered such as moving averages and volume ratios. These variables help capture changes in trading activity and track broader trends in prices.

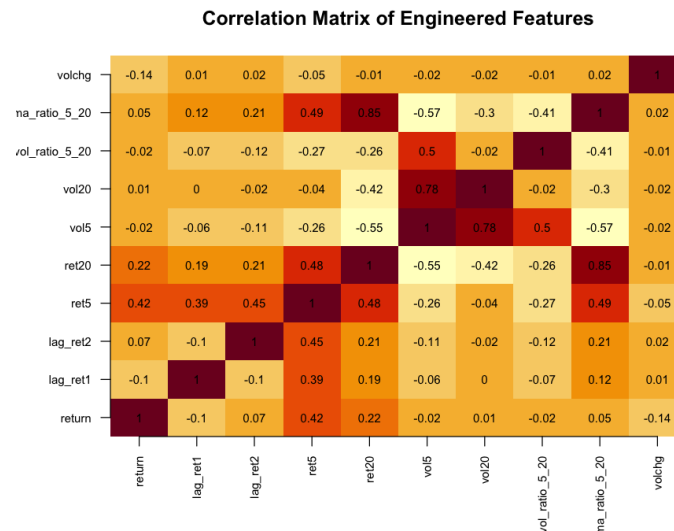


Figure 4: Feature Correlation Matrix

Most of the engineered features have little to no correlation with each other which suggests that the feature set contains a good mix of different information. However, a few features are highly correlated

with each other, `ret20` is highly correlated with `ma_ratio.5_20`, with a correlation of about 0.85. This makes sense because both variables capture medium-term momentum or trend strength. `ret5` has moderate positive correlations with `return`, `lag_ret1`, `lag_ret2`, and `ret20`, showing that short-term returns are somewhat connected across nearby time periods.

5 Statistical Methods

5.1 Gaussian Processes

Gaussian Processes are defined by the equation (Roberts et al., 2013)

$$f(x) \sim GP(m(x), k(x, x')),$$

where $m(x)$ represents the mean function and $k(x, x')$ represents the covariance function. The covariance between two points in time measures how similar those two points are based on their inputs (Roberts et al., 2013). In this case, the input is time. In general, points that are closer together in time are expected to have a stronger relationship, while points that are farther apart are expected to have a weaker relationship. Gaussian Processes use this idea by calculating the covariance between every pair of time points and organizing those values into a covariance matrix. If we define the set of time inputs as

$$x = \{x_1, x_2, x_3, x_4, \dots, x_n\}$$

and define the covariance operation using the kernel function k , then the covariance matrix can be written as

$$K(x, x) = \begin{pmatrix} k(x_1, x_1) & k(x_1, x_2) & \cdots & k(x_1, x_n) \\ k(x_2, x_1) & k(x_2, x_2) & \cdots & k(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ k(x_n, x_1) & k(x_n, x_2) & \cdots & k(x_n, x_n) \end{pmatrix}.$$

Each entry in this matrix represents the covariance between two time points, allowing the Gaussian Process to model how observations depend on one another across time. The mean function represents the expected value of the function at each input x . In a Gaussian Process, this is written as

$$\mu(x) = \begin{bmatrix} m(x_1) \\ m(x_2) \\ \vdots \\ m(x_n) \end{bmatrix}$$

This means that $m(x)$ describes the average or expected behavior of the unknown function before observing the data. In the context of financial time series data, the mean function can be used to represent the expected movement of returns. The next step to complete the Gaussian Process is to apply a normal distribution to the matrix of the mean function and covariance function,

$$\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_m) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(x_1) \\ \vdots \\ m(x_m) \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_m) \\ \vdots & \ddots & \vdots \\ k(x_m, x_1) & \cdots & k(x_m, x_m) \end{bmatrix} \right).$$

Applying a multivariate normal distribution is necessary because it allows the Gaussian Process to model many points in the time series at the same time. The model assumes that the collection of function values across the time period follow a joint Gaussian distribution. The normal distribution is closed under linear transformations and conditioning, this means that if returns are assumed to be normally distributed, then linear transformations of those returns will also be normally distributed. Conditioning in terms of financial returns means that the model uses past observed returns to update its beliefs about future returns. The Gaussian Process begins with a joint distribution over both historical and future returns. After observing the historical VOO returns, the model conditions on those values to form a new distribution for the unknown future returns. Since the normal distribution is closed under conditioning, this updated predictive distribution is still normal. This property allows the model to estimate future returns while also measuring the uncertainty around those estimates.

5.2 Bayesian Additive Regression Trees

A Bayesian Additive Regression Tree (BART) model is a non-parametric, decision tree-based learning method used to model the relationship between a response variable y and a set of predictors x (Chipman et al., 2010). Rather than assuming a fixed linear relationship between the predictors and the response, BART approximates the unknown function by adding together many small regression trees. Each individual tree explains only a small portion of the relationship in the data, but when all of the trees are combined, they form a flexible and powerful predictive model.

BART is similar to other tree based learning models such as Random Forest, bagging and gradient boosting. They all combine smaller trees to create strong predictive model. Instead of relying on one individual decision tree, these methods use an ensemble of trees, where each tree captures a different part of the relationship between the predictors and the response variable. This is useful because a single tree can be unstable or overly sensitive to the data, while a collection of trees can produce more accurate and flexible predictions. However, a key difference between is that BART introduces a regularization prior. BART modeling inference is conducted under a Bayesian framework by placing priors over tree structures, terminal node parameters, and the error variance (Chipman et al., 2010). BART generates a posterior distribution over many possible functions which allows the model to estimate predicted return and the uncertainty surrounding the prediction.

In a tree-based model, each individual tree partitions the predictor space through a sequence of binary decision rules. For example, a tree may split the data based on whether yesterday's return was positive or negative. Each split sends observations down different branches of the tree until they reach a terminal node. Each terminal node is assigned a parameter value, which represents the predicted contribution of that tree for observations that fall into that region of the predictor space.

For a model with a single regression tree, the response variable can be written as

$$Y = g(x; T, M) + \varepsilon$$

where Y represents the response variable, x represents the set of predictors, T represents the tree structure, M represents the set of terminal node parameters, and ε represents the error term. The error term in this instance is capturing what is not explained by the regression tree and in the next equation the sum of trees model. The unexplained could be random noise, unobserved factors or anything the trees fail to capture. When the model is expanded to more trees the model is represented as this, giving it the name “sum of trees”;

$$Y = \sum_{j=1}^m g(x; T_j, M_j) + \varepsilon$$

Each observation represents a specific trading day. BART creates the final predictive model by passing each observation through every tree, where it follows a sequence of binary decision rules until it reaches a terminal node. The terminal node represents the final endpoint of a branch where no other splits will occur. The value assigned to that terminal node represents that tree's contribution to the prediction. After this process is repeated across all trees, BART adds the individual tree contributions together to produce the final predicted value.

6 Application

6.1 Setup

A convenient way to fit a Gaussian Process to a dataset in R is using the **GauPro** package (Erickson, 2025). The **gpkm** function is the primary function used from the package to run the Gaussian Process (Erickson, 2025). This function has a few inputs which allow us to control the GP. The first input X represents the matrix of input predictors used to explain variation in the response variable. Each row of X corresponds to one trading day, while each column represents an engineered financial feature, such as lagged returns, rolling volatility, moving averages, or volume-based variables. This input is used to train the GP using a data split which varies for each period of targeted returns. The input Z is the response variable being predicted, for this project this input will be returns. Those are the two main inputs in the **gpkm** function however there are a few more features that we need to designate. The

inputs `normalize` and `parallel` are both set to `FALSE`, if `normalize` is set to `TRUE` it would normalize our response variable but we already normalized returns so this input is not necessary. `parallel` tells the model to use parallel computing, this is set to `FALSE` as it helps us avoid the potential errors that come with splitting up the computations. Both `verbose` and `restarts` are set to 0, `verbose` just controls how much the model prints while running. Setting it to 0 helps improve run times. `restarts` controls how many times the optimizer restarts when the GP is estimating parameters, this keeps the model fitting process simpler, but more restarts could potentially improve the parameter estimates.

For the BART model, R provides a package called `dbarts`, which works similarly to other decision tree modeling packages (Dorie et al., 2026). This package allows us to fit a Bayesian Additive Regression Tree model based on our inputs (Dorie et al., 2026). For `dbarts` we must designate our `x.train` and `y.train` matrices, `x.train` is our predictor matrix which contains our engineered features. `y.train` is our response matrix which consists of some variation of returns depending on the prediction. Next we define our testing predictor matrix this is designated by `x.test`. This matrix contains the same predictors as `x.train` however it contains the observations not used in the training model. Next we define how large we want our tree to be, `ntree`, `ndpost` and `nskip` are the model settings for this. The parameter `ntree` determines the number of regression trees used in the BART model. The parameter `ndpost` represents the number of posterior draws saved after the burn-in period. The parameter `nskip` represents the number of initial Monte Carlo Markov Chain samples that are discarded before inference begins. The BART model was fit using 50 trees. Increasing the number of trees can improve model flexibility because it allows the model to capture more complex relationships between the engineered financial predictors and returns. However, after testing models with a larger number of trees, there was not a meaningful improvement in predictive performance. Since financial return data is highly noisy, using too many trees can increase the risk of overfitting random short-term movements rather than learning useful patterns. For this reason, a smaller tree ensemble was preferred because it helps avoid unnecessary model complexity while also improving run time.

7 Results and Discussion

7.1 Daily Returns

The first period of returns tested by both the GP and BART model is daily returns. This consists of 252 predictions made sequentially for 2025, 252 represents the number of trading days in 2025.

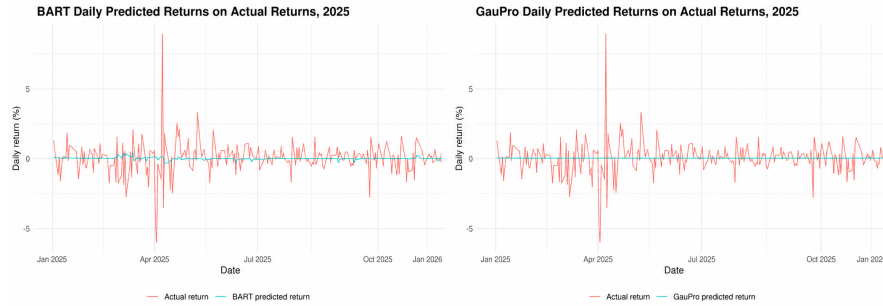


Figure 5: VOO Daily Returns

Daily returns prove far too difficult to predict because of how volatile they are. Daily price movements are influenced by random market movements, information and investor emotions. These make it extremely difficult to model short-term returns suggesting that these values contain a lot of noise. The S&P 500 moved on average 0.73% in 2025, both the GP model and BART model had a RMSE around 1.5%. Comparing this to predicting a 0% the models underperformed. This indicates that, for daily returns, the models were often making prediction errors larger than the typical daily movement itself. Daily market movements are essentially a flip of a coin, the results support this as daily return markets are dominated by random fluctuations driven by information, making them too difficult to predict even with our advanced modeling techniques.

7.2 Quarterly Returns

The next period analyzed are quarterly returns, widening the period predicted by the BART and GP models. Each financial quarter is made up of 63 trading days widening the period of returns models 63x.

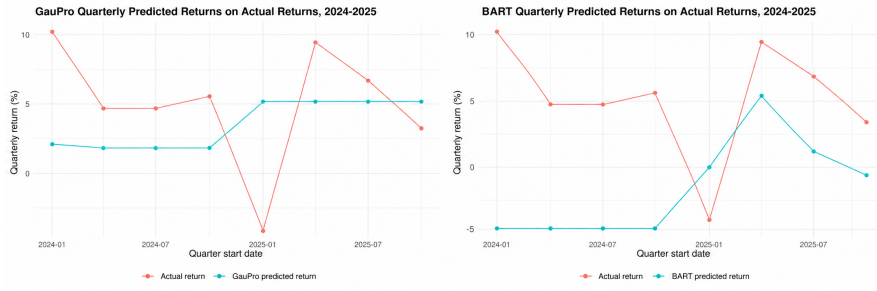


Figure 6: VOO Quarterly Returns

This longer return horizon helped smooth out some of the short-term noise that dominated the daily data. Quarterly returns gave the models a better opportunity to capture broader trends, momentum, and changes in market conditions over time. As a result, the predictions became more meaningful than the daily return predictions.

7.3 Yearly Returns

The final period examined in this project was yearly returns, which represented the longest prediction horizon tested in the analysis. The first time frame analyzed for model comparison was 2022-2025.

Table 1: BART Model Quarterly Return Predictions

Year	Actual	Predicted	Error	RMSE
2022	-22.00	15.57	37.57	0.0137
2023	20.46	11.11	9.35	0.0186
2024	21.38	9.47	11.91	0.0186
2025	16.87	10.42	6.45	0.0186

Table 2: GP Model Yearly Return Predictions

Year	Actual	Predicted	Error	RMSE
2022	-22.00	32.76	54.75	0.0333
2023	20.46	4.55	15.90	0.2070
2024	21.38	-1.66	23.04	0.1124
2025	16.87	19.97	3.10	0.0120

The BART and GP models modeled yearly returns far better than the daily and quarterly models. The general direction of the movement in returns is far better captured and shows some initial signs of applicability in the financial field. The daily and quarterly return models did not distinguish a clear winner between the GP and BART models. However, the BART model sports on average a much lower RMSE than the GP model and it captures the overall movement in yearly returns. Therefore, the next step was to apply the BART model to a longer yearly return period from 2017 to 2025. This larger window allows for a better evaluation of the BART and to see if it can consistently capture return patterns in different market conditions.

7.4 Expanded Yearly Returns

Table 3: Expanded BART Model Yearly Return Predictions

Year	Actual	Predicted	Error	RMSE
2017	18.42	10.46	7.98	0.0186
2018	-9.59	11.91	21.50	0.0173
2019	26.05	12.00	14.04	0.0186
2020	14.12	10.52	3.60	0.0186
2021	25.90	10.91	14.99	0.0196
2022	-22.00	15.57	37.57	0.0137
2023	20.46	11.11	9.35	0.0186
2024	21.38	9.47	11.91	0.0186
2025	16.87	10.42	6.45	0.0186

On the expanded yearly return window from 2017 to 2025, the BART model showed a more consistent ability to capture the general level of long-term returns compared with the shorter-horizon models. The predictions were relatively stable from year to year, usually falling around the 10 to 15 percent range, which suggests that the model was identifying a broad average yearly return pattern. However, the model struggled during more extreme years, especially 2018 and 2022, where actual returns were negative but the model still predicted positive returns. This can be explained by market information that was not included in the dataset. In 2022 to combat a 40-year high in inflation the Federal Reserve raised interest rates 11 consecutive times starting in March 2022. Energy prices in 2022 surged as well due to the war between Russia and Ukraine breaking out. Information such as these two events were not included in the dataset, which helps explain why the BART model struggled to predict the sharp decline in returns during that year. It was trained on historical stock prices and failed to account for major macroeconomic shocks or geopolitical events.

7.5 Model Comparison

Table 4: Model Performance Comparison Across Return Frequencies

Model	Freq	Obs	RMSE	MAE	ME
GP	Daily	250	1.159	0.724	0.016
BART	Daily	250	1.170	0.741	-0.060
Linear	Daily	250	1.232	0.765	0.006
BART	Overall	262	3.189	1.200	-0.233
Linear	Overall	262	3.835	1.356	-0.279
GP	Overall	262	4.068	1.192	0.041
GP	Quarterly	8	5.097	4.330	-1.517
BART	Quarterly	8	8.947	7.998	-7.003
Linear	Quarterly	8	10.524	9.098	-6.009
BART	Yearly	4	20.507	16.319	2.464
Linear	Yearly	4	25.436	22.815	-6.657
GP	Yearly	4	30.787	24.199	4.728

The model comparison table provides gives us a better view of how the models performed over the different horizons when compared to each other. A linear regression model was added to provide a baseline model. For daily returns, the three models performed very similarly, with the GP model producing the lowest RMSE at 1.159, followed closely by BART at 1.170 and the linear model at 1.232. The small difference between these values suggests that the more advanced Bayesian models did not create a major improvement over the linear model at the daily level. For quarterly returns, the GP model performed best across the main error metrics. It had the lowest RMSE at 5.097 and the lowest MAE at 4.330, while BART had an RMSE of 8.947 and the linear model had an RMSE of 10.524. The yearly return comparison shows a different pattern. BART had the lowest RMSE at

20.507 and the lowest MAE at 16.319, while the GP model had the highest RMSE at 30.787 and MAE at 24.199. This means that BART was the strongest model at the yearly level in the four-year testing window. The linear model was between the two Bayesian models, with an RMSE of 25.436 and MAE of 22.815. When the data is evaluated across the overall testing period, BART performs best by RMSE with a value of 3.189, compared to 3.835 for the linear model and 4.068 for the GP model. Overall, the comparison table shows that model performance depends heavily on the return frequency being predicted. GP performed best for quarterly returns, while BART performed best for the overall and yearly evaluations. This suggests that there is no single model that dominates across every time horizon, but BART appears to be the most useful model for longer-term return prediction in this project.

8 Conclusion

This paper examined whether Gaussian Processes and Bayesian Additive Regression Trees can improve stock return prediction when applied to VOO market data. The results show that short-term returns are far too volatile to accurately predict. Daily returns were heavily affected by random market movement, noise, and sudden changes in investor behavior. Because of this, all models struggled with daily return predictions, including the more complex Bayesian models. This supports the idea that stock data is historically unpredictable, and that accurately predicting market movement is something even professionals struggle with.

The results also showed that model performance depends heavily on the time horizon being predicted. Direction and magnitude were captured much better over longer return periods than over short-term daily returns. The quarterly and yearly models gave the GP and BART models more room to identify broader market trends instead of reacting to daily noise. In general, the Bayesian models tended to outperform the linear model, especially when predicting longer-term returns. However, the results also show that more complex models are not always automatically better. In some cases, the linear model performed similarly to the Bayesian models, which suggests that a simple model can sometimes be more reliable when the data is noisy or when the added complexity does not create a major improvement.

Overall, BART appeared to be the strongest model for longer-term return prediction, while Gaussian Processes performed well in certain quarterly comparisons. However, both models still had important shortcomings. Even with a small number of trees, BART could have long run times, and increasing the model size did not always lead to better performance. This suggests that future versions of the model should focus on shrinking the model and avoiding unnecessary complexity. Future research could also expand the dataset by incorporating quarterly predictors, earnings data, financial ratios, and other variables that capture information raw prices cannot. Adding derivatives data, such as S&P 500 futures, may also improve the model because futures can act as leading indicators of market expectations. Finally, future work should compare these models with other decision tree methods, Bayesian models, and stronger baseline models to better evaluate whether Bayesian methods provide a consistent advantage in financial prediction. While these models do not solve the problem of market unpredictability, they provide a useful framework for studying uncertainty, nonlinear relationships, and longer-term return patterns in financial data.

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